

Aswanth Bindu Ajith

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1 PROFESSIONAL SUMMARY

M.Sc. student in High Performance Computing and Quantum Computing with a Mechanical Engineering foundation. Comfortable building high-performance computing, computational simulation, software engineering solutions from low-level numerical code to production web features. Presented optimization findings to a 25-person seminar and received top marks. Reduced simulation runtime by 40

2 TECHNICAL SKILLS

HPC Parallel Computing: MPI (basic), OpenMP (basic), GPU-aware computing concepts, Cluster job scheduling (SLURM basics), Performance profiling

Quantum Computing: Qiskit, Cirq (basic), Quantum optimization algorithms, Variational quantum eigen-solver (VQE)

Machine Learning: PyTorch, Scikit-learn, TensorFlow (basic), Feature engineering, Model evaluation

Modeling Simulation: Finite difference methods, Computational fluid dynamics (basics), Thermal simulation, Multi-physics modeling

3 PROFESSIONAL EXPERIENCE

Working Student Software Developer	Germany
No Border (E-commerce)	Feb 2025 Present
- Shipped 5+ frontend features in a 2-week sprint cycle, reducing average ticket age by 30 - Debugged and resolved 20+ production bugs across the React/Vue stack, cutting reported error rates by 25 - Wrote unit tests for 3 core modules, pushing test coverage from 45 - Collaborated with 4-person cross-functional team (design, backend, QA) using GitHub Flow and daily standups	

4 EDUCATION

M.Sc. High Performance Computing and Quantum Computing	2024 Present
Deggendorf Institute of Technology (DIT), Germany	Current Average: 2.6
<i>Relevant coursework:</i> Parallel Distributed Computing, Numerical Optimization, Quantum Algorithms, Scientific Software Engineering, High-Dimensional Data Analysis	
<i>Highlights:</i> Optimization Methods (1.7); System Design Applications (1.7); Software Engineering (2.3); Advanced Mathematics (2.3)	
B.Tech Mechanical Engineering	2019 2023
APJ Abdul Kalam Technological University, India	CGPA: 6.48

5 PROJECTS

Computing Architecture Optimization Methods
- Implemented gradient descent and genetic algorithms from scratch to optimize a 50-dimensional engineering cost function - Benchmarked CPU vs. GPU (CUDA) performance on matrix operations, documenting 8x speedup for large matrices - Profiled memory bottlenecks using Python's cProfile and reduced peak RAM usage by 35 - Presented findings to a 25-person seminar, receiving top marks on the technical report - Key metrics: 50-dimensional optimization, 8x GPU speedup, 35
Large-Scale Computational Simulation Data Analysis
- Built a Python simulation engine processing 30,000+ interacting entities using vectorized NumPy operations - Reduced runtime from 12 minutes to 7 minutes (40

- Identified 3 non-obvious data correlations that shifted the team's modeling assumptions
- Packaged results into an 18-page LaTeX report with reproducible Jupyter notebooks for peer review
- Key metrics: 30,000+ entities, 40

Thermal Analysis of Battery Modules

- Modeled heat transfer across a 12-cell lithium-ion battery pack using finite difference methods in Python
- Simulated 8 operating scenarios, identifying a thermal runaway risk at $>45^{\circ}\text{C}$ under sustained 3C discharge
- Proposed a revised cooling channel geometry that improved peak temperature uniformity by 18
- Co-authored a technical report cited by 2 subsequent student projects in the same lab
- Key metrics: 12-cell pack, 8 scenarios, 18

6RESEARCH INTERESTS

High Performance Computing, Performance Optimization, GPU Computing, Scientific Software Engineering

7LANGUAGES

English (Professional Working Proficiency), Malayalam (Native), German (A1 (Beginner))